

3,000 alerts. One analyst.

Utopia SOC

Transparent grouping. Human decisions. Measured coverage.

The analyst needs an investigation, not another wall of signals.

CONNECT

Bring entity and time evidence together across identity, endpoint, and network alerts.

PRIORITIZE

Expose asset context, behavior progression, and each risk component.

VERIFY

Measure how many attack episodes become visible for a given review budget.

Security decisions stay deterministic.

Validate → deduplicate → correlate → ATT&CK map → classify → score → rank

Every source alert is retained. Every accepted link has a reason.

Simulation truth → evaluation only. No labels in engine inputs.

Facts packet → verified template / optional model brief.

Analyst decision → versioned state + SHA-256 audit chain.

Raw events → detectors → validated alerts.

FICTIONAL ENTERPRISE

120 hosts · 160 users · 72-hour alert batch. Causal state prerequisites and seeded raw events.

FOUR ATTACK FAMILIES

Phishing / exfiltration · identity compromise · ransomware · command and control.

DIFFICULTY CONTROLS

Sensor dropout · duplicates · missing entities · shared hubs · concurrency · timing and severity noise.

A queue that opens into evidence.

utopia

SECURITY OPERATIONS

Aster Financial

Private browser workspace

WORKSPACE

Import reports

Benchmark lab

Incident queue 69

Raw alerts

Evaluation

Simulation lab

Audit trail

How it works

Analyst in control

Evidence guides decisions.

No autonomous response.

Imagine Utopia

Team 239

Workspace > Incident queue

EVIDENCE WORKBENCH | UTC IU

TRIAGE WORKSPACE

Incident queue

Turn alert noise into investigations worth opening.

JSON

Report

CSV

Run simulation

Seed 239

RUN-d329f7f9-62d · 3,000 alerts

Delete analysis

3,000 accepted · 0 source rows rejected

Deterministic engine v2.0

Incoming alerts

3,000

Across EDR, identity & network.

Review items

617

79.4% fewer queue rows

Critical & high

69

1 critical · 68 high priority

Pipeline runtime

115 ms

Measured on this run

From signals to investigations

This batch

3,000

Source alerts

DEDUPLICATE

977

Distinct groups

CORRELATE

617

Review items

Queue compression is measured. Time savings require analyst testing.

Signal sources

3 connected schemas

Endpoint detection

1,622

Identity provider

713

Network detection

665

Ranked Investigations

617

Highest risk first · deterministic tie-breaks

Search incidents, hosts, users...

All risk levels

All statuses

RISK	INCIDENT	ENTITIES	ALERTS	FIRST SEEN	STATUS
82 Critical	Potential ransomware activity INC-B19FF5B3 · 5 mapped behaviors	2 hosts 2 users	18	13:16 28 Sept	Open
80 High	Credential compromise with lateral movement	2 hosts	16	07:25	Open

Live workflow: ranked queue → source timeline → graph reasons → score → human disposition

05

Open the relationship. Inspect the reason.

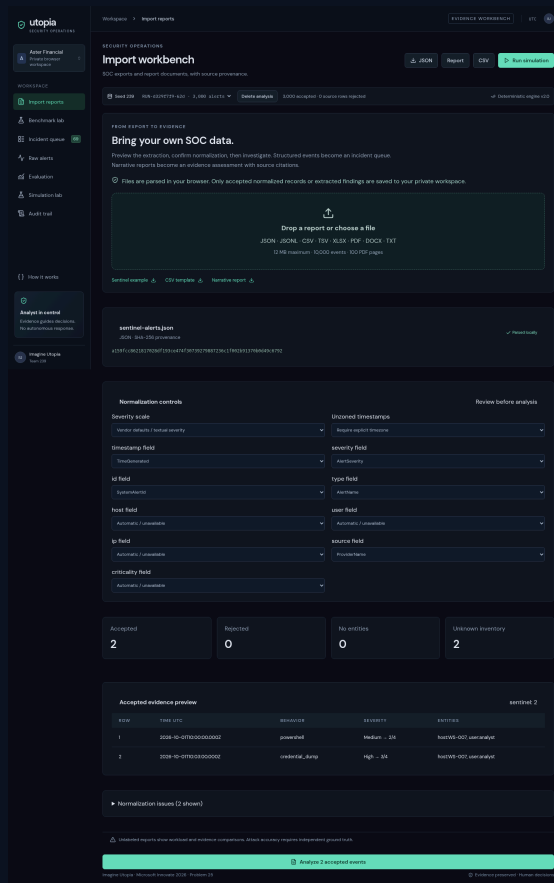
The screenshot displays a security dashboard with a sidebar on the left containing navigation links like 'Incident queue', 'Alerts', and 'Investigations'. The main area shows an 'Incident queue' with a table of incidents. One incident is highlighted, showing a 'Potential ransomware activity' with a 'Critical' severity and '18 source alerts'. The incident is dated '13:16 - 13:31 UTC' and is currently 'Open'.

The detailed view for this incident shows a 'Correlation' tab. It features a graph with six nodes (G1, G2, G3, G4, G5, G6) representing different alert groups. The nodes are connected by lines, indicating relationships. The nodes are labeled as follows:

- G1: powershell
- G2: dns beacon
- G3: suspicious login
- G4: scheduled task
- G5: powershell
- G6: encryption activity

The graph shows a complex network of connections between these alert groups. A green line highlights a specific correlation link between G1 and G2. Below the graph, a status bar indicates 'Accepted correlation link' and 'Threshold passed'. At the bottom, there are three buttons: 'Mark false positive', 'Investigate', and 'Escalate'.

Bring reports. Keep the evidence.



Same inputs. Same labels. Same metric code.

B0 Severity only — one alert per review item

B1 Host / 30-minute buckets

B2 Exact typed-entity sets / time buckets

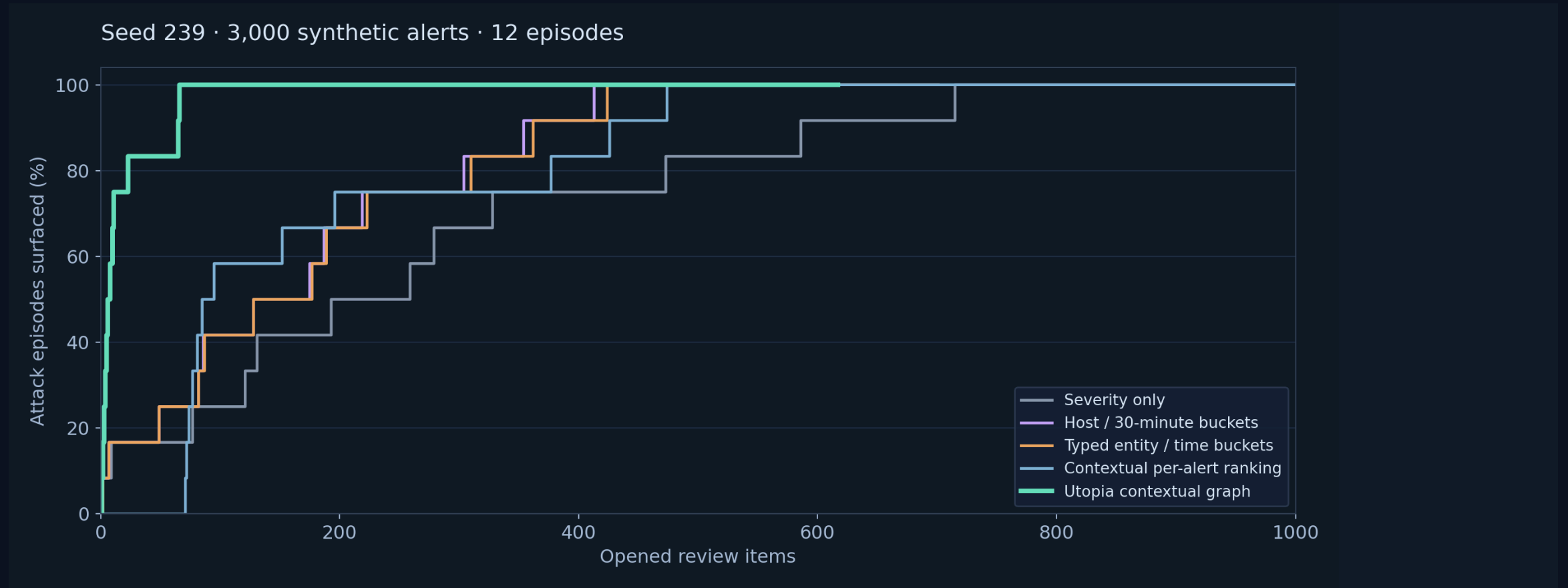
B3 Contextual per-alert ranking

UT Contextual entity graph + deterministic score

Episode surfaced = at least one of its alerts in an opened item.

Pairwise grouping quality separately measures reconstruction.

3,000 alerts → 617 review items.



Twenty held-out seeds at each scale.

80.0% RECALL @25

Utopia mean across held-out
3,000-alert runs. B0 mean: 17%.

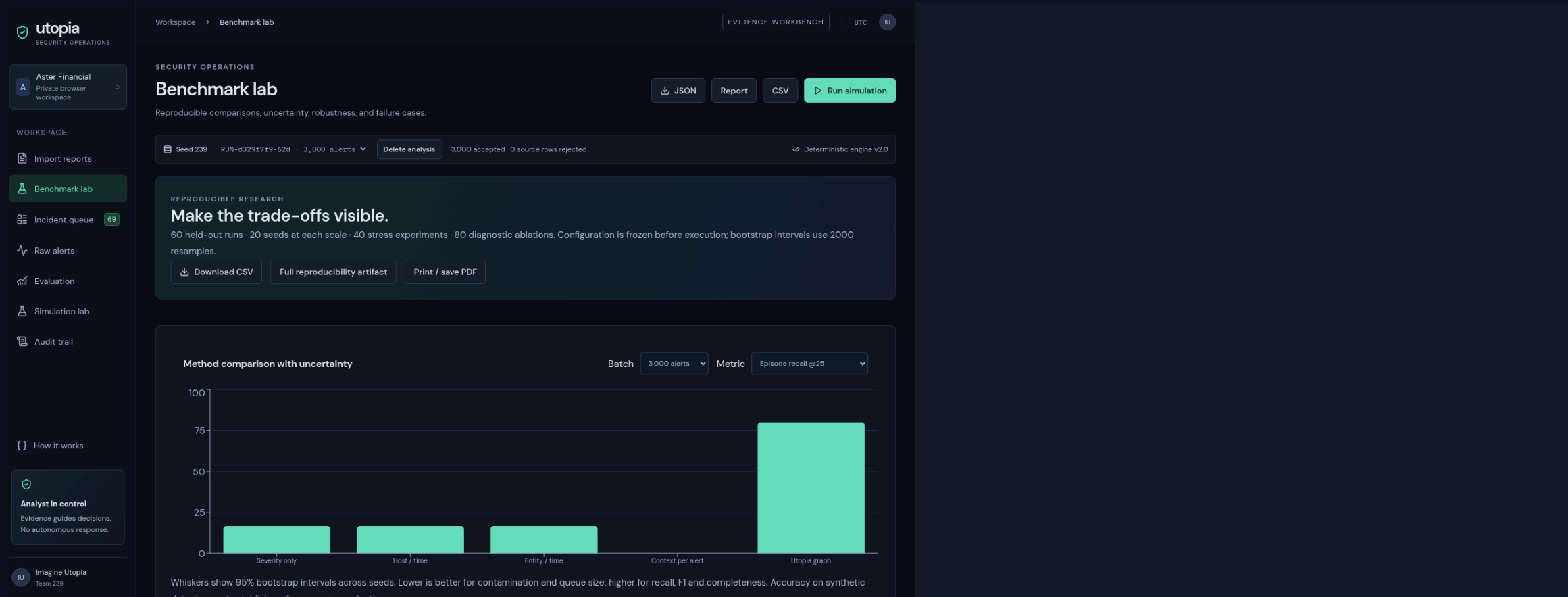
54.9 ITEMS TO FULL COVERAGE

Utopia mean. Severity-only:
589.8 items. Total Utopia queue:
624.8.

25.6% GROUPING F1

Mean benign contamination:
3.7%. Surfacing is not full
reconstruction.

Show uncertainty, not just a winning seed.



Long gaps fragment the chain.

80-MINUTE STEP GAPS

Seed 801, 3,000 alerts. Frozen entity windows no longer connect many stages.

16.7% RECALL @25

Pairwise recall: 17.8%. Full coverage takes 267 items.

NEXT EXPERIMENT

Measure wider windows, contamination, and segmentation on separate tuning data; never hide the trade-off.

Complete workflow. Honest boundaries.

Implemented: simulation, ingest, correlation, evidence, scoring, dashboard, decisions, audit, baselines, benchmarks.

Live brief: deterministic template; optional model adapter is in source.

AIT-ADS check: checksum verified; 3,000 Wazuh records accepted / 400 AMiner rejected. No gold attack accuracy inferred.

Audit: trusted-head tamper evidence; no external anchoring.

No autonomous response, compliance guarantee, or production accuracy claim.

Make the trade-off visible.

Can transparent grouping surface attack episodes with fewer opened review items?

Synthetic evidence: improved episode visibility, with declining recall and modest grouping F1 as scale increases.

Next: independently labeled external data, a real analyst study, and measured review effort.